A Two-Phase Integer L-Shaped Method for Stochastic Electric-Bus Wireless Charging Infrastructure Planning

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Abstract

Dynamic wireless charging (DWC) can reduce electric-bus battery requirements by supplying energy in transit. Yet its long-lived infrastructure must be planned before future service plans are known. This study investigates how to coordinate DWC facility locations with scenario-dependent route-level battery configurations and representative-day charging schedules. We formulate the problem as a two-stage stochastic program that minimizes facility, battery, and charging costs. Each scenario specifies fixed timetables and fleet requirements. Direct use of the integer L-shaped method requires repeatedly solving large mixed-integer scenario problems in which shared plug-in capacity couples routes. A weak master approximation can also trigger expensive evaluations of layouts that later turn out infeasible or costly. To solve the problem, we develop a repair-enhanced two-phase integer L-shaped method (R-TPILS). An equivalent reformulation lifts battery choices into the master and supports energy-budget inequalities. In Phase 1, relaxing shared capacity decomposes each scenario into route-level linear programs that generate lower bounds and cuts. Phase 2 restores shared capacity and verifies candidates against the exact scenario problems. A repair search exploits feasibility monotonicity to seek verified feasible solutions and strengthen feasibility cuts. Computational experiments on the Tower Transit bus network in Singapore compare R-TPILS with the extensive formulation and three integer L-shaped alternatives. R-TPILS meets the 1% gap target on all nine main instances, compared with at most eight for any alternative. In these experiments, the energy-budget inequalities screen infeasible configurations early, and Phase 1 quickly establishes tight lower bounds. On challenging instances, R-TPILS also finds verified feasible solutions where the no-repair version does not. The planning study shows that DWC reduces mean battery capacity by about 23% and shifts energy supply from overnight toward daytime charging.

Keywords: wireless charging facility location, electric buses, two-stage stochastic optimization, integer L-shaped method

1. Introduction

Electric buses are being adopted to reduce local air pollution and noise, but their limited driving range may require recharging during the day (de Vos et al., 2024). Dynamic wireless charging (DWC) can supplement plug-in charging by supplying energy to vehicles while they are moving. DWC is particularly well suited to bus systems, where fixed routes and frequent services allow the same charging infrastructure to serve many buses repeatedly.

DWC deployment involves substantial upfront investment in infrastructure intended to remain in use for many years. During this period, changes in passenger demand may require adjustments to bus service plans (Perumal et al., 2022). The deployment decision must therefore account for how the facilities will be used under uncertain future service plans. We study this decision in a bus network with given plug-in charger locations and capacities. The DWC layout is chosen before the service plan is known. Once a service-plan scenario is realized, the planner selects a route-level battery configuration for that planning period and a representative-day charging schedule. Because routes share the existing plug-in stations, their charging schedules must jointly satisfy station capacity limits. The resulting two-stage stochastic DWC planning model is a mixed-integer program. With all costs expressed on a representative-day basis, its objective minimizes DWC facility cost and the expected costs of battery and charging.

The two-stage structure suggests an integer L-shaped (ILS) decomposition, with DWC decisions in the master and battery and charging decisions in scenario subproblems. A direct application faces two computational difficulties. The master represents the feasibility and recourse cost of a DWC layout through cuts. A weak approximation can lead to repeated evaluations of layouts whose infeasibility or high recourse cost is revealed only after subproblem solution. At the same time, each scenario subproblem remains a large mixed-integer linear program (MILP) after scenario decomposition. Battery and charger-occupancy decisions are discrete, while shared plug-in charging capacity couples the routes within the subproblem. Repeated exact solution of integer recourse can be expensive (Larsen et al., 2024), magnifying the cost of a weak master approximation. These difficulties motivate a method that strengthens the master's representation of recourse and obtains useful bounds before exact scenario evaluation.

Our contributions are threefold.

i. We formulate a two-stage stochastic DWC planning model that coordinates long-term DWC deployment with scenario-dependent route battery configurations and representative-day charging schedules. We characterize its recourse structure, in which shared capacity at the existing plug-in stations is the sole source of coupling between routes.
ii. To solve the model, we derive an equivalent reformulation and use it to develop a two-phase integer L-shaped method (TPILS) that exploits the route separability revealed when shared

plug-in capacity constraints are relaxed. We further develop a repair-enhanced variant (R-TPILS) that uses feasibility monotonicity to strengthen the decomposition.

iii. We conduct numerical experiments based on a 28-route Tower Transit network in Singapore to evaluate the proposed methods using the extensive formulation (EF) and ILS as baselines. A planning case study compares DWC-enabled and no-DWC plans in terms of battery capacity and charging allocation.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 introduces the planning problem and formulates the two-stage stochastic DWC planning model. Section 4 presents the equivalent reformulation, TPILS, and R-TPILS. Section 5 reports the computational experiments and planning comparisons. Finally, Section 6 concludes with the main findings and future research directions.

2. Literature Review

Two streams of research inform this study: electric-bus charging facility location and decomposition methods for stochastic programs with mixed-integer recourse.

2.1. Electric-bus charging facility location

For a given bus service, larger batteries can reduce the need for daytime charging, while more charging opportunities at stations can allow buses to operate with smaller batteries (Perumal et al., 2022). Daytime charging must fit around scheduled trips, making both travel to charging stations and waiting for a charger relevant to facility planning. Bazarnovi et al. (2027) jointly choose station locations and the number of chargers at each station under stochastic charging demand, balancing investment against the time costs of travel to stations, waiting, and charging. By supplying energy as buses travel, DWC adds charging opportunities along a route. The locations and lengths of equipped road segments affect how much energy a bus can receive and, in turn, the battery capacity needed to serve the route. In a system with existing plug-in chargers at depots and terminals, Li et al. (2024) jointly determine DWC deployment, route-level battery capacities, and charging schedules under time-of-use electricity prices.

The operating conditions that determine charging needs can be uncertain. Li et al. (2021) use a two-stage stochastic model to locate refueling facilities and schedule a mixed bus fleet before the driving ranges of electric buses are known. Additional services are deployed after the ranges are realized to cover scheduled trips left incomplete because of energy shortages. Under uncertain travel times and energy consumption, Wang et al. (2024) use a two-stage robust model to determine the locations and lengths of DWC facilities and the battery capacities for each route. En-route charging is adjusted after the uncertain conditions are realized. The service plan itself may also be

uncertain when facilities are deployed. Different plans may specify different trips and assign them to buses in different ways, changing which DWC-equipped links buses traverse and when they can charge at stations.

For a given service plan, buses on different routes may need to charge at the same station, where the number of chargers limits how many buses can charge at once. Given the trips assigned to each bus, Abdelwahed et al. (2020) optimize the timing and duration of charging subject to these limits. Other studies jointly assign trips to buses and schedule charging, subject to station capacity limits (Gkiotsalitis et al., 2023; de Vos et al., 2024; Ricard et al., 2027). Even if a feasible charging schedule exists for each route separately, there may be no feasible schedule for all routes together because buses from different routes compete for the same chargers.

2.2. Decomposition methods for stochastic integer programs

Benders decomposition separates an optimization problem into a master problem and subproblems, using cuts from the subproblems to refine the master. In classical Benders decomposition, the subproblems are linear programs (LPs) whose dual solutions and rays provide optimality and feasibility cuts. For two-stage stochastic programs with continuous linear recourse, the L-shaped method applies this decomposition across scenarios. With integer recourse, an LP relaxation may underestimate the recourse cost or fail to detect infeasibility. For problems with binary first-stage decisions and complete recourse, the integer L-shaped method uses integer recourse evaluations to construct cuts (Laporte and Louveaux, 1993). Logic-based Benders decomposition also accommodates integer subproblems, deriving cuts through problem-specific reasoning (Hooker and Ottosson, 2003). Nevertheless, limited information from cuts can lead to many iterations, while solving the subproblems at each iteration can be costly (Rahmaniani et al., 2017).

Reformulations can make recourse information available to the master before cuts are generated. Retaining selected scenario subproblems in the master provides an explicit representation of their recourse decisions (Crainic et al., 2021; Çelik et al., 2025). Moving individual scenario-dependent variables into the master follows the same principle, but can make the master large when there are many scenarios. Lai et al. (2025) exploits customer choices that coincide across scenarios to reduce this growth. Even with a stronger master, evaluating recourse across many scenarios can make cut generation expensive. Aggregating scenarios or sampling a subset can reduce the work involved (Ramírez-Pico et al., 2023; Bertsimas et al., 2025). The aggregation can be refined using subproblem dual solutions, whereas sampled evaluations can be combined through dual averaging to construct valid cuts.

Problem structure can provide additional information for the master through valid inequalities and stronger cuts. Li et al. (2022) use processing and release times to bound the makespan and strengthen cuts by identifying smaller sets of assignments that cannot improve the incumbent. Sim-

ilarly, feasibility cuts can be strengthened by removing assignments and re-solving the subproblem to identify a smaller set that remains infeasible (Elçi and Hooker, 2022). For mixed-integer recourse, lower approximations of recourse costs can also support optimality cuts (van der Laan and Romeijnders, 2024; Baals et al., 2026). Monotonicity is another property that can extend the information obtained from a subproblem evaluation to other master solutions. Examples include cuts based on monotone route recourse costs and on simulation evaluations of resource allocations (Parada et al., 2024; Forbes et al., 2024).

When subproblems are expensive to solve, relaxations can provide cheap information to the master. LP relaxations of integer recourse, for example, can generate cuts and eliminate some candidate solutions before integer subproblems are solved (Angulo et al., 2016; Satici and Dayarian, 2026). Çalık and Fortz (2019) solve a partially relaxed model and check whether its optimal value can be attained by an integer solution with the same first-stage decisions. If not, they continue the search and add combinatorial feasibility cuts for integer-infeasible subproblems. Relaxing coupling constraints can also simplify the master problem, but a feasible solution must ultimately satisfy those constraints. Chen and Liu (2023) relax the nonanticipativity of the initial vehicle allocation, allowing each scenario to choose its own allocation, and also relax vehicle-flow integrality. They then restore a common allocation and use a diving heuristic to construct integer operational solutions. Thus, relaxed solutions can support both cut generation and feasible-solution search, while the constraints and integrality requirements of the original problem must still be addressed.

Existing studies have developed reformulations, valid inequalities, strengthened cuts, and relaxation-based procedures to improve Benders decomposition, as summarized in Table 1. In our problem, fixing the DWC deployment leaves a recourse problem in each scenario that selects battery capacities and charging schedules. The limited capacity of plug-in stations couples charging decisions across routes, so each scenario subproblem remains a large mixed-integer program. In the natural decomposition, the master does not explicitly represent these battery and charging decisions and must learn their feasibility and cost through cuts. Repeatedly solving the coupled scenario subproblems can therefore be computationally expensive. These difficulties motivate a decomposition method that strengthens the master's representation of recourse and exploits route structure to reduce the burden of scenario evaluation.

Table 1: Selected components of decomposition methods

Study	Original subproblem	Model reformulation	Valid inequalities	Monotonicity usage	Relaxation	Relaxation postprocessing
Angulo et al. (2016)	MILP				Integrality	Cuts
Çalık and Fortz (2019)	ILP	✓	✓		Integrality	Cuts
Crainic et al. (2021)	LP	✓	✓		Integrality	Cuts
Chen and Liu (2023)	ILP				Integrality; constraints	Cuts; repair
Ramírez-Pico et al. (2023)	LP	✓			—	—
Lai et al. (2025)	MILP	✓		✓	Constraints	Cuts
Çelik et al. (2025)	LP	✓			—	—
Bertsimas et al. (2025)	QP	✓	✓		Integrality	Cuts
Baals et al. (2026)	MILP	✓	✓		Constraints; objective	Cuts
Satici and Dayarian (2026)	MILP	✓			Integrality; constraints	Cuts
This paper	MILP	✓	✓	✓	Integrality; constraints	Cuts; repair

3. Problem Setting and Formulation

This section defines the directed station network and its scheduled traversal events and formulates the DWC facility location problem as a two-stage stochastic program.

3.1. Problem description

Consider an electric bus system with a set of stations $\mathcal{S}$ and a set of directed links $\mathcal{L}$. A link connects consecutive stations along a bus duty, including links traversed between consecutive trips. Existing plug-in chargers are available at stations in $C \subseteq \mathcal{S}$. These chargers have given capacities and are shared by buses in the system. DWC facilities may be installed on candidate links in $\mathcal{L}^{\mathrm{D}} \subseteq \mathcal{L}$.

The system operates a set of routes $\mathcal{R}$. For a given service plan, each bus duty consists of an ordered sequence of trips assigned to one bus and any travel between consecutive trips. A bus departs from the depot with its battery at the upper state-of-charge limit and then operates its assigned trips in sequence. As illustrated in Figure 1, during its duty the bus may use plug-in charging and receive DWC while traversing equipped links. After completing its last trip, the bus returns to the depot and is recharged overnight to the upper state-of-charge limit before the next operating day. Depot charging capacity is sufficient to recharge all buses overnight.

We consider alternative service plans for a single planning period. As shown in Figure 2, the planner first selects the links on which DWC facilities are installed, before the service plan is known. For each possible service plan, the planner then chooses a matching route-level battery

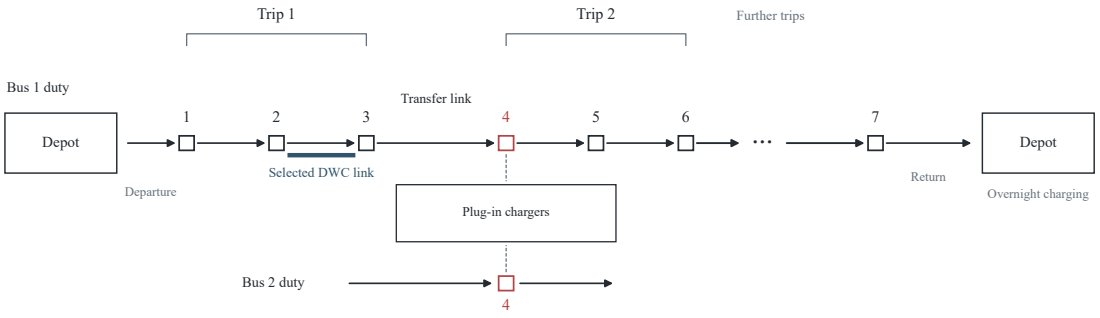


Figure 1: Illustration of a bus duty, its charging opportunities, and a plug-in station shared with other buses.

configuration and representative-day charging plan under the same facility layout. The service plan and battery configuration remain fixed throughout the planning period.

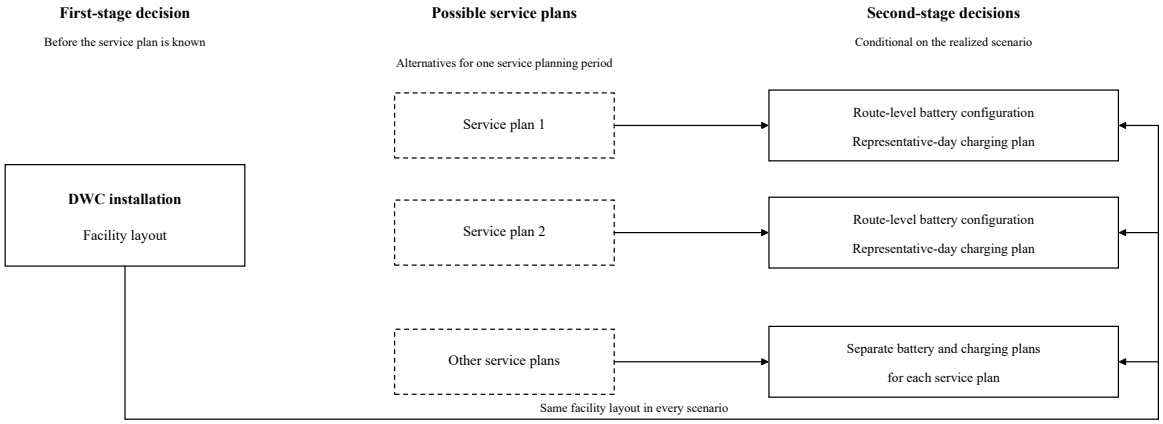


Figure 2: Two-stage decisions under alternative service-plan scenarios.

3.2. Two-stage stochastic formulation

3.2.1. Overall formulation and first-stage decisions

The possible service plans are represented by scenarios $\omega \in \Omega$, each with probability $p_\omega > 0$, where $\sum_{\omega\in\Omega} p_\omega = 1$. All costs are expressed on a representative-day basis, and the notation is summarized in Appendix A. $C_\omega(u_\omega)$ represents the second-stage cost for scenario ω. The binary variable w_l indicates whether DWC is installed on candidate link l, and u_ω represents the battery and charging decisions for scenario ω. The model minimizes the DWC facility cost and the expected

battery and charging costs:

$$\begin{aligned} \min_{w,\{u_\omega\}_{\omega\in\Omega}} \quad & C^{\text{fac}}(w) + \sum_{\omega\in\Omega} p_\omega C_\omega(u_\omega) \\ \text{s.t.} \quad & w \in \mathcal{W}, \\ & u_\omega \in \mathcal{U}_\omega(w), \qquad \omega \in \Omega. \end{aligned} \tag{1}$$

In (1), $\mathcal{W}$ is the first-stage feasible set, and $\mathcal{U}_\omega(w)$ is the second-stage feasible set under scenario ω and facility layout w. The first-stage feasible set and facility cost are

$$\begin{aligned} \mathcal{W} &= \{0,1\}^{|\mathcal{L}^{\text{D}}|}, \\ C^{\text{fac}}(w) &= \sum_{l\in\mathcal{L}^{\text{D}}} c_l^{\text{D}} w_l, \end{aligned} \tag{2}$$

where c_l^{D} is the cost of installing DWC on link l.

3.2.2. Second-stage decisions and constraints

For scenario ω, write $u_\omega = (z_\omega, x_\omega, y_\omega)$, where z_ω collects route-level battery selections, x_ω plug-in charger occupancy decisions, and y_ω charging amounts and battery energy states. The scenario cost is $C_\omega(u_\omega) = C_\omega^{\text{bat}}(z_\omega) + C_\omega^{\text{chg}}(y_\omega)$. The following constraints define $\mathcal{U}_\omega(w)$ for each scenario $\omega \in \Omega$.

Battery configuration and cost. Battery option $k \in \mathcal{K}$ has capacity β_k, and the scenario-dependent binary variable $z_{\omega,r,k}$ indicates whether it is selected for route r. The route-level battery-choice constraints are

$$\sum_{k\in\mathcal{K}} z_{\omega,r,k} = 1, \qquad r \in \mathcal{R}, \tag{3}$$

$$B_{\omega,r} = \sum_{k\in\mathcal{K}} \beta_k z_{\omega,r,k}, \qquad r \in \mathcal{R}, \tag{4}$$

$$z_{\omega,r,k} \in \{0,1\}, \qquad r \in \mathcal{R},\ k \in \mathcal{K}. \tag{5}$$

Constraints (3)–(5) select one common battery option for all buses assigned to each route and define the induced route battery capacity $B_{\omega,r}$. For the set $\mathcal{M}_{\omega,r}$ of bus duties on route r, $|\mathcal{M}_{\omega,r}|$ is the number of buses. Let c^{bat} denote the battery capacity cost per kWh. The scenario battery cost is

$$C_\omega^{\text{bat}}(z_\omega) = c^{\text{bat}} \sum_{r\in\mathcal{R}} |\mathcal{M}_{\omega,r}| \sum_{k\in\mathcal{K}} \beta_k z_{\omega,r,k}.$$

This term gives the battery capacity cost for all buses required under the service plan; the acquisition or disposal of vehicles and batteries associated with changes in fleet size is not modeled separately.

Across the service plans, $\mathcal{V}$ and $\mathcal{A}$ denote the sets of station-visit and link-traversal events generated by bus duties, respectively. Each duty is represented by an ordered sequence of station visits as shown in Figure 1. Each visit $v \in \mathcal{V}$ is a distinct occurrence in a duty, and $s(v)$ is its physical station. Each duty, visit, and traversal belongs to a single scenario–route pair. Each traversal $a = (v, v') \in \mathcal{A}$ connects consecutive visits on the same duty and is associated with the physical link $\ell(a) = (s(v), s(v'))$. As shown in Figure 3, the same physical link may be traversed multiple times by the same bus or by different buses, with each traversal represented by a distinct event.

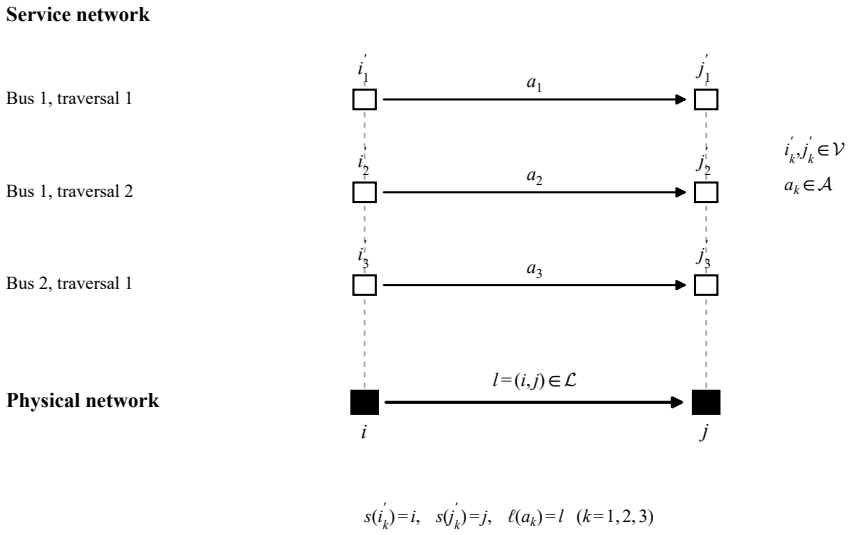


Figure 3: Mapping from service-network traversal events to a physical link. The three rows represent distinct traversals; dashed lines associate visit events with their physical stations.

Shared plug-in capacity. The station visits on route r in scenario ω are denoted by $\mathcal{V}_{\omega,r}$, and $\mathcal{V}^{\mathrm{P}}_{\omega,r}$ contains the terminal visits eligible for plug-in charging under the service plan. An eligible visit v can charge during its available time slots $\mathcal{T}_v \subseteq \mathcal{T}_\omega$, where $\mathcal{T}_\omega$ contains the relevant time slots in scenario ω. The variable q^{P}_v denotes the grid-side energy supplied and $x_{v\tau}$ indicates whether a visit

occupies a charger in slot τ. The plug-in charging constraints are

$$\begin{cases} 0 \le q_v^{\mathrm{P}} \le \bar{q}^{\mathrm{P}} \sum_{\tau \in \mathcal{T}_v} x_{v\tau}, & r \in \mathcal{R},\ v \in \mathcal{V}^{\mathrm{P}}_{\omega,r}, \\ q_v^{\mathrm{P}} = 0, & r \in \mathcal{R},\ v \in \mathcal{V}_{\omega,r} \setminus \mathcal{V}^{\mathrm{P}}_{\omega,r}, \end{cases} \tag{6}$$

$$\sum_{r \in \mathcal{R}} \sum_{\substack{v \in \mathcal{V}^{\mathrm{P}}_{\omega,r}: \\ s(v)=c,\ \tau \in \mathcal{T}_v}} x_{v\tau} \le N_c, \qquad c \in C,\ \tau \in \mathcal{T}_\omega, \tag{7}$$

$$x_{v\tau} \in \{0,1\}, \qquad r \in \mathcal{R},\ v \in \mathcal{V}^{\mathrm{P}}_{\omega,r},\ \tau \in \mathcal{T}_v. \tag{8}$$

Constraint (6) limits grid-side energy delivery to $\bar{q}^{\mathrm{P}}$ per occupied slot and excludes charging at ineligible visits. Constraint (7) limits simultaneous charger occupancy at station c to the number N_c of existing chargers and couples buses using the same station. Constraint (8) specifies binary occupancy decisions.

Other operating constraints and charging cost. Let $\mathcal{A}_{\omega,r}$ denote the link traversals on route r in scenario ω, and $\mathcal{A}^{\mathrm{D}}_{\omega,r}$ the subset on candidate DWC links. For a traversal a, q_a^{D} denotes the grid-side DWC energy supplied; for a duty $m \in \mathcal{M}_{\omega,r}$, q_m^{N} denotes the grid-side overnight energy supplied. The DWC energy bound $\bar{q}_a^{\mathrm{D}}$ is the maximum amount of energy that can be supplied via DWC for a given traversal. The battery energy levels upon arrival at and departure from visit v are denoted by e_v^{arr} and e_v^{dep}, respectively. For duty m, v_m^0 and v_m^N denote its first and last station visits, and d_m^0 and d_m^N denote the energy consumed traveling from and to the depot, respectively. The consumption d_a is assumed to be known and fixed for each traversal a, and η^{P} and η^{D} convert grid-side charging energy to battery energy. The charging and energy variables in y_ω are nonnegative. For each route $r \in \mathcal{R}$, the operating constraints are

$$\begin{cases} 0 \le q_a^{\mathrm{D}} \le \bar{q}_a^{\mathrm{D}} w_{\ell(a)}, & a \in \mathcal{A}^{\mathrm{D}}_{\omega,r}, \\ q_a^{\mathrm{D}} = 0, & a \in \mathcal{A}_{\omega,r} \setminus \mathcal{A}^{\mathrm{D}}_{\omega,r}, \end{cases} \tag{9}$$

$$\underline{\rho} B_{\omega,r} \le e_v^{\mathrm{arr}} \le \overline{\rho} B_{\omega,r}, \qquad v \in \mathcal{V}_{\omega,r}, \tag{10}$$

$$\underline{\rho} B_{\omega,r} \le e_v^{\mathrm{dep}} \le \overline{\rho} B_{\omega,r}, \qquad v \in \mathcal{V}_{\omega,r}, \tag{11}$$

$$e_v^{\mathrm{dep}} = e_v^{\mathrm{arr}} + \eta^{\mathrm{P}} q_v^{\mathrm{P}}, \qquad v \in \mathcal{V}_{\omega,r}, \tag{12}$$

$$e_{v'}^{\mathrm{arr}} = e_v^{\mathrm{dep}} - d_a + \eta^{\mathrm{D}} q_a^{\mathrm{D}}, \qquad a = (v, v') \in \mathcal{A}_{\omega,r}, \tag{13}$$

$$e_{v_m^0}^{\mathrm{arr}} = \overline{\rho} B_{\omega,r} - d_m^0, \qquad m \in \mathcal{M}_{\omega,r}, \tag{14}$$

$$e_{v_m^N}^{\mathrm{dep}} - d_m^N \ge \underline{\rho} B_{\omega,r}, \qquad m \in \mathcal{M}_{\omega,r}, \tag{15}$$

$$\eta^{\mathrm{P}} q_m^{\mathrm{N}} = \overline{\rho} B_{\omega,r} - e_{v_m^N}^{\mathrm{dep}} + d_m^N, \qquad m \in \mathcal{M}_{\omega,r}. \tag{16}$$

Constraint (9) links DWC charging to facility decisions, and constraints (10)–(11) impose the state-of-charge limits $\underline{\rho}$ and $\overline{\rho}$. Constraints (12)–(16) describe energy balances and depot operations for each duty.

At daytime and overnight electricity prices $c^{\text{day}} \geq 0$ and $c^{\text{night}} \geq 0$, the scenario charging cost is

$$C_\omega^{\text{chg}}(y_\omega) = c^{\text{day}} \left(\sum_{r \in \mathcal{R}} \sum_{v \in \mathcal{V}^{\text{P}}_{\omega,r}} q_v^{\text{P}} + \sum_{r \in \mathcal{R}} \sum_{a \in \mathcal{A}^{\text{D}}_{\omega,r}} q_a^{\text{D}} \right) + c^{\text{night}} \sum_{r \in \mathcal{R}} \sum_{m \in \mathcal{M}_{\omega,r}} q_m^{\text{N}}.$$

4. A Two-Phase Integer L-Shaped Method

The two-stage structure of (1) naturally lends itself to an integer L-shaped decomposition (Laporte and Louveaux, 1993). The master determines the scenario-independent facility layout w, and each scenario subproblem selects batteries and charging operations under that layout. We refer to this decomposition as Natural ILS. With $\Phi_\omega(w) = \min\{C_\omega(u_\omega) : u_\omega \in \mathcal{U}_\omega(w)\}$ denoting the scenario value, the exact epigraph formulation is

$$\begin{aligned} \min_{w,\theta^{\text{N}}} \quad & C^{\text{fac}}(w) + \sum_{\omega \in \Omega} p_\omega \theta_\omega^{\text{N}} \\ \text{s.t.} \quad & w \in \mathcal{W}, \\ & \theta_\omega^{\text{N}} \geq \Phi_\omega(w), \qquad \omega \in \Omega. \end{aligned} \tag{17}$$

Here $\Phi_\omega(w) = +\infty$ if the scenario subproblem is infeasible, and θ_ω^{N} represents its combined battery and charging cost.

4.1. Variable lifting and valid inequalities

In Natural ILS, battery capacities and their costs are determined within the scenario subproblems and enter the master through $\Phi_\omega(w)$. We lift the battery-selection variables z_ω into the master to represent the trade-off between shared facility investment and scenario-dependent battery capacity explicitly. The resulting master jointly selects facilities and battery configurations across scenarios, accounting directly for battery costs and selection constraints.

The lifted master is

$$\begin{aligned} V^L = \min_{w,z,\theta} \quad & C^{\text{fac}}(w) + \sum_{\omega \in \Omega} p_\omega [C_\omega^{\text{bat}}(z_\omega) + \theta_\omega] \\ \text{s.t.} \quad & w \in \mathcal{W}, \\ & (3)\text{–}(5), \qquad r \in \mathcal{R},\ \omega \in \Omega, \\ & \theta_\omega \geq Q_\omega(w, z_\omega), \qquad \omega \in \Omega. \end{aligned} \tag{18}$$

Here $z = (z_\omega)_{\omega\in\Omega}$ and θ_ω represents scenario charging cost. For fixed (w, z_ω), the operational subproblem is

$$\begin{aligned} Q_\omega(w, z_\omega) = \min_{x_\omega, y_\omega} \quad & C^{\mathrm{chg}}_\omega(y_\omega) \\ \text{s.t.} \quad & (6)\text{–}(8), \\ & (9)\text{–}(16), \quad r \in \mathcal{R}, \\ & y_\omega \geq 0. \end{aligned} \tag{19}$$

Set $Q_\omega(w, z_\omega) = +\infty$ if the subproblem is infeasible. Together, these problems retain all variables, costs, and constraints of (1). We refer to ILS applied directly to this formulation as Lifted ILS.

Energy-budget valid inequality.. For each $m \in \mathcal{M}_{\omega,r}$, let $\mathcal{A}_m$ contain its traversals, and let $\mathcal{V}^{\mathrm{P}}_m$ and $\mathcal{A}^{\mathrm{D}}_m$ contain its eligible plug-in visits and DWC traversals, respectively. Usable battery energy and maximum charging supply must cover total duty consumption, yielding the following energy-budget valid inequality (VI):

$$\begin{aligned} &(\overline{\rho} - \underline{\rho}) \sum_{k\in\mathcal{K}} \beta_k z_{\omega,r,k} + \eta^{\mathrm{P}} \bar{q}^{\mathrm{P}} \sum_{v\in\mathcal{V}^{\mathrm{P}}_m} |\mathcal{T}_v| \\ &\quad + \eta^{\mathrm{D}} \sum_{a\in\mathcal{A}^{\mathrm{D}}_m} \bar{q}^{\mathrm{D}}_a w_{\ell(a)} \geq d^0_m + \sum_{a\in\mathcal{A}_m} d_a + d^N_m, \qquad \forall\, \omega \in \Omega,\ r \in \mathcal{R},\ m \in \mathcal{M}_{\omega,r}. \end{aligned} \tag{20}$$

We add the inequality for the duty with the largest number of trips in each scenario–route pair to the master in both phases.

4.2. Two-phase relaxation and verification

As the road network and fleet expand, the operational subproblem (19) grows in both charging decisions and shared-capacity constraints, making it more difficult to solve. To address this computational burden, we develop TPILS consisting of two phases. In Phase 1, removing the shared-capacity coupling allows occupancy variables to be eliminated, decomposing each large scenario MILP into small, independent route LPs. Phase 2 restores the exact scenario MILPs while retaining the lower bounds generated in Phase 1. Figure 4 summarizes the two phases.

Phase 1: Coupling relaxation and route decomposition.. For fixed (w, z_ω), we drop the shared plug-in capacity constraints (7) and eliminate occupancy variables, obtaining

$$0 \leq q^{\mathrm{P}}_v \leq \bar{q}^{\mathrm{P}} |\mathcal{T}_v|, \qquad v \in \mathcal{V}^{\mathrm{P}}_{\omega,r}. \tag{21}$$

After the shared-capacity constraints are removed, any charging amount satisfying this bound can be represented by setting $x_{v\tau} = 1$ for every $\tau \in \mathcal{T}_v$. The remaining constraints involve one route at

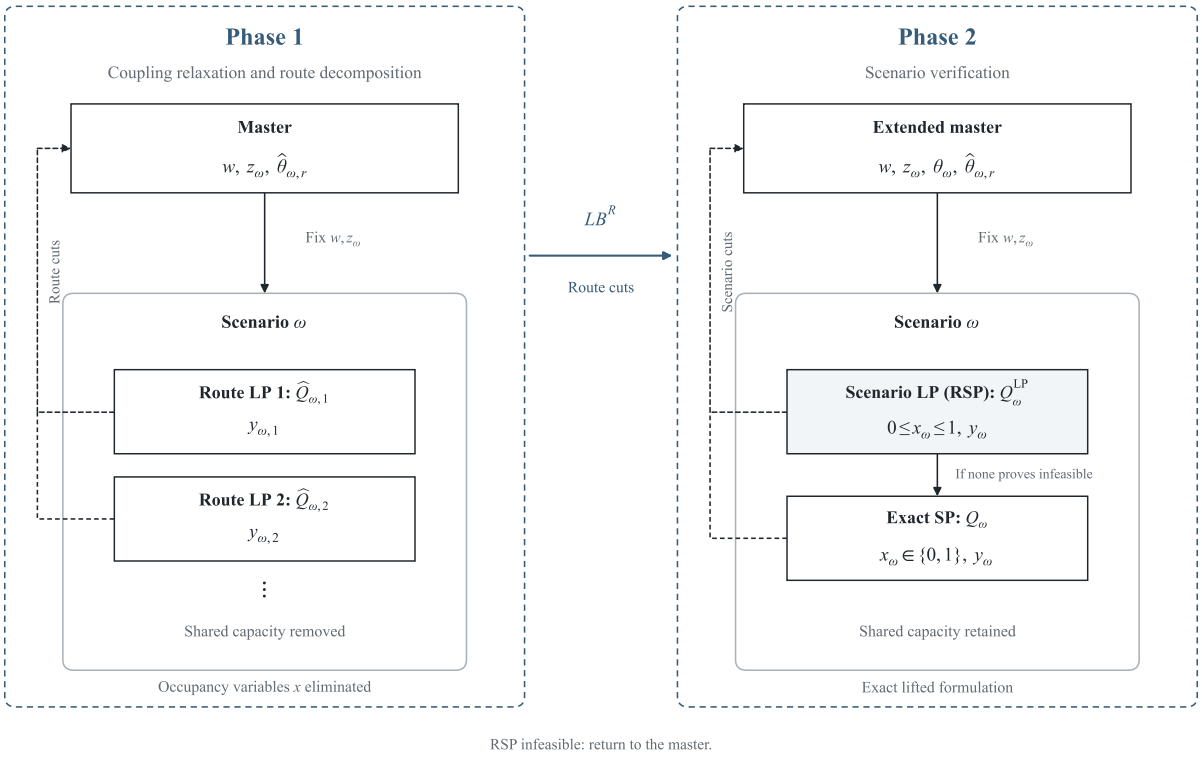


Figure 4: Two-phase solution framework for TPILS.

a time, and charging cost is additive across routes, $C^{\mathrm{chg}}_\omega = \sum_{r\in\mathcal{R}} C^{\mathrm{chg}}_{\omega,r}$. For each (ω, r), the resulting route LP is

$$\begin{aligned} \widehat{Q}_{\omega,r}(w, z_{\omega,r}) = \min_{y_{\omega,r}\ge 0} \quad & C^{\mathrm{chg}}_{\omega,r}(y_{\omega,r}) \\ \text{s.t.} \quad & (9)\text{–}(16), \\ & (21), \\ & q^{\mathrm{P}}_v = 0, \qquad v \in \mathcal{V}_{\omega,r} \setminus \mathcal{V}^{\mathrm{P}}_{\omega,r}. \end{aligned} \tag{22}$$

All constraints are restricted to route r. Set $\widehat{Q}_{\omega,r}(w, z_{\omega,r}) = +\infty$ if the route LP is infeasible.

Proposition 1 (Validity of the route relaxation)**.** *For any facility layout $w \in \mathcal{W}$, battery selections satisfying* (3)–(5)*, and $\omega \in \Omega$,*

$$\sum_{r\in\mathcal{R}} \widehat{Q}_{\omega,r}(w, z_{\omega,r}) \le Q_\omega(w, z_\omega). \tag{23}$$

Infeasibility of any route LP certifies infeasibility of the exact scenario subproblem.

Proof. Since $x_{v\tau} \le 1$, every exact feasible solution satisfies (21) and retains all other route constraints. Its route components are therefore feasible for the route LPs with the same total charging

cost, proving the lower bound and, by contraposition, the infeasibility claim. If the exact subproblem is infeasible, the bound holds by $Q_\omega = +\infty$. □

The Phase 1 restricted master is the following program augmented with all accumulated route feasibility and optimality cuts:

$$\begin{aligned}
\min_{w,z,\widehat{\theta}} \quad & \left[C^{\text{fac}}(w) + \sum_{\omega\in\Omega} p_\omega C^{\text{bat}}_\omega(z_\omega)\right] + \sum_{\omega\in\Omega} p_\omega \sum_{r\in\mathcal{R}} \widehat{\theta}_{\omega,r} \\
\text{s.t.} \quad & w \in \mathcal{W}, \\
& (3)\text{–}(5), && \omega \in \Omega,\ r \in \mathcal{R}, \\
& (20), \\
& \widehat{\theta}_{\omega,r} \geq 0, && \omega \in \Omega,\ r \in \mathcal{R}.
\end{aligned} \tag{24}$$

The bracketed term is the facility and battery cost of (w, z); the remaining term approximates route charging costs through proxies $\widehat{\theta}_{\omega,r}$ constrained by the accumulated route cuts. Optimal route LP solutions generate dual optimality cuts; infeasible route LPs generate feasibility cuts, using a Farkas certificate when available (Benders, 1962). Each cut involves only its route's battery selection and facilities.

Let LB^R be the best Phase 1 master bound and UB^R the lowest evaluated objective for a design with all route LPs feasible. Each evaluation adds facility and battery costs to the expected charging cost obtained from the route LPs. The phase transition occurs when

$$g^R = \frac{UB^R - LB^R}{\max\{1, |UB^R|\}} \leq \varepsilon_1. \tag{25}$$

Set $g^R = +\infty$ until UB^R is finite. The value LB^R also bounds V^L, whereas UB^R bounds only the relaxed problem.

Phase 2: Exact scenario verification.. Phase 2 extends the Phase 1 master to solve the exact lifted formulation (18), carrying forward the lower bound LB^R and route cuts. In the objective of (24), the sum of the route cost proxies $\sum_{r\in\mathcal{R}} \widehat{\theta}_{\omega,r}$ for each scenario is replaced by the scenario cost proxy θ_ω. Its restricted master approximates the scenario value constraints $\theta_\omega \geq Q_\omega(w, z_\omega)$ with accumulated feasibility and optimality cuts. The selected energy-budget inequalities are retained, and the route and scenario cost proxies are linked through

$$\begin{aligned}
\theta_\omega &\geq \sum_{r\in\mathcal{R}} \widehat{\theta}_{\omega,r}, && \omega \in \Omega, \\
\widehat{\theta}_{\omega,r} &\geq 0, && \omega \in \Omega,\ r \in \mathcal{R}.
\end{aligned} \tag{26}$$

At each Phase 2 master solution (w^*, z^*, θ^*), we first solve one relaxed scenario subproblem (RSP) for every $\omega \in \Omega$. Let $Q_\omega^{\mathrm{LP}}(w, z_\omega)$ denote the value of the linear relaxation of (19), obtained by replacing (8) with $0 \le x_\omega \le 1$; set $Q_\omega^{\mathrm{LP}} = +\infty$ if the relaxation is infeasible. The RSP retains shared plug-in capacity across routes and therefore captures interactions omitted by the route LPs. Its value satisfies $\sum_{r\in\mathcal{R}} \widehat{Q}_{\omega,r} \le Q_\omega^{\mathrm{LP}} \le Q_\omega$. In our computational experience, finite RSP values are typically close to their exact counterparts, yielding strong lower-bounding cuts for the master. As shown on the right of Figure 4, we solve the exact scenario MILPs only if all RSPs are feasible; otherwise, we return to the master. Both RSPs and exact scenario MILPs provide optimality cuts when solved to optimality and feasibility cuts when infeasible. RSP cuts use LP dual information when available; exact scenario MILPs use integer L-shaped optimality cuts (Laporte and Louveaux, 1993) and no-good feasibility cuts. Once a master design is feasible for every exact scenario subproblem, its evaluated cost provides an upper bound. TPILS terminates when this upper bound and the restricted-master lower bound satisfy the prescribed relative tolerance.

4.3. Monotone feasibility cuts and repair enhancement

R-TPILS augments TPILS with the repair-enhanced monotone feasibility cuts developed below. When the exact subproblem for scenario ω certifies (w^*, z_ω^*) as infeasible, TPILS adds the following pointwise no-good feasibility cut (Elçi and Hooker, 2022):

$$D_\omega(w, z_\omega; w^*, z_\omega^*) = \sum_{l\in\mathcal{L}^{\mathrm{D}}: w_l^*=0} w_l + \sum_{l\in\mathcal{L}^{\mathrm{D}}: w_l^*=1} (1 - w_l) + \sum_{r\in\mathcal{R}} \sum_{k\in\mathcal{K}} z^*_{\omega,r,k}(1 - z_{\omega,r,k}) \ge 1. \tag{27}$$

On the binary domain, this cut excludes only the evaluated facility and battery configuration.

Monotone feasibility cuts.. Operational feasibility is monotone with respect to DWC availability and battery capacity. If (w^*, z_ω^*) is infeasible, any configuration with no additional facilities and no larger batteries is also infeasible. A feasible configuration must therefore open an additional DWC facility or increase the battery capacity on at least one route, giving the monotone feasibility cut

$$I_\omega(w, z_\omega; w^*, z_\omega^*) = \sum_{l\in\mathcal{L}^{\mathrm{D}}: w_l^*=0} w_l + \sum_{r\in\mathcal{R}} \sum_{k\in\mathcal{K}: \beta_k > B_{\omega,r}(z^*_{\omega,r})} z_{\omega,r,k} \ge 1. \tag{28}$$

Proposition 2 (Validity and dominance)**.** *Cut* (28) *is valid and weakly dominates* (27) *at the same anchor.*

Proof. First, consider a feasible configuration (w, z_ω) and another configuration $(\bar{w}, \bar{z}_\omega)$ with $\bar{w}_l \ge w_l$ for every $l \in \mathcal{L}^{\mathrm{D}}$ and $B_{\omega,r}(\bar{z}_{\omega,r}) \ge B_{\omega,r}(z_{\omega,r})$ for every $r \in \mathcal{R}$. Keeping the original charging and occupancy decisions, the DWC charging bounds remain satisfied because $\bar{w} \ge w$, while the plug-in charging and capacity constraints are unaffected. For each route, write $B_{\omega,r} = B_{\omega,r}(z_{\omega,r})$,

$\bar{B}_{\omega,r} = B_{\omega,r}(\bar{z}_{\omega,r})$, and $\Delta_r = \bar{B}_{\omega,r} - B_{\omega,r} \geq 0$; increase every arrival and departure energy state by $\overline{\rho}\Delta_r$. The common shift preserves station and traversal energy balances throughout each duty. For every original state e, its original bounds imply

$$e + \overline{\rho}\Delta_r \geq \underline{\rho} B_{\omega,r} + \overline{\rho}\Delta_r \geq \underline{\rho}\bar{B}_{\omega,r},$$
$$e + \overline{\rho}\Delta_r \leq \overline{\rho} B_{\omega,r} + \overline{\rho}\Delta_r = \overline{\rho}\bar{B}_{\omega,r}.$$

The second inequality in the first line follows from $\overline{\rho} \geq \underline{\rho}$ and $\Delta_r \geq 0$. The initial boundary equality is preserved; return energy increases by $\overline{\rho}\Delta_r$, at least the increase $\underline{\rho}\Delta_r$ in its lower bound. In the overnight balance, the increases in the capacity term and terminal energy cancel, so the original overnight charging remains valid. The transformed solution is feasible at $(\bar{w}, \bar{z}_\omega)$.

Consequently, a configuration with no additional facilities and no larger batteries than (w^*, z^*_ω) must be infeasible. On the binary domain, these configurations are exactly those with $I_\omega = 0$, proving validity.

By the battery-selection constraint (3), the battery terms in I_ω are no greater than their counterparts in D_ω; the same holds for the facility terms. Thus $I_\omega \leq D_\omega$ on the continuous relaxation, so (28) weakly dominates (27). □

Repair-enhanced monotone cuts.. Because the objective minimizes cost, the master may generate infeasible configurations close to previously evaluated anchors. The resulting monotone cuts may therefore provide only limited additional coverage of the infeasible region. Starting from (w^*, z^*_ω), the repair search first upgrades batteries and then adds facilities, seeking a feasible configuration and identifying resource-enlarged configurations that remain infeasible. The last such infeasible configuration found, denoted by (w^+, z^+_ω), is used as the anchor of the strengthened cut:

$$\begin{aligned} I_\omega(w, z_\omega; w^+, z^+_\omega) \geq 1, \qquad & Q_\omega(w^+, z^+_\omega) = +\infty, & \\ w^+_l \geq w^*_l, & & l \in \mathcal{L}^{\mathrm{D}}, \\ B_{\omega,r}(z^+_{\omega,r}) \geq B_{\omega,r}(z^*_{\omega,r}), & & r \in \mathcal{R}. \end{aligned} \tag{29}$$

This cut can exclude a larger region of infeasible configurations.

Figure 5 illustrates the search for one scenario. Battery upgrades are listed in ascending order of incremental cost, and unbuilt facilities in ascending order of construction cost. A prefix of length h contains the h lowest-cost entries in the corresponding list. The first evaluation upgrades each eligible route's battery by one level (point 1). Because this candidate is feasible, the search evaluates successively halved prefixes of the battery-upgrade list until it finds the first infeasible candidate (point 2). Starting from point 2, the facility search applies the same prefix-halving procedure to the facility list and stops when it finds the infeasible anchor at point 4. The additional

shaded region shows the increase in coverage relative to the original cut.

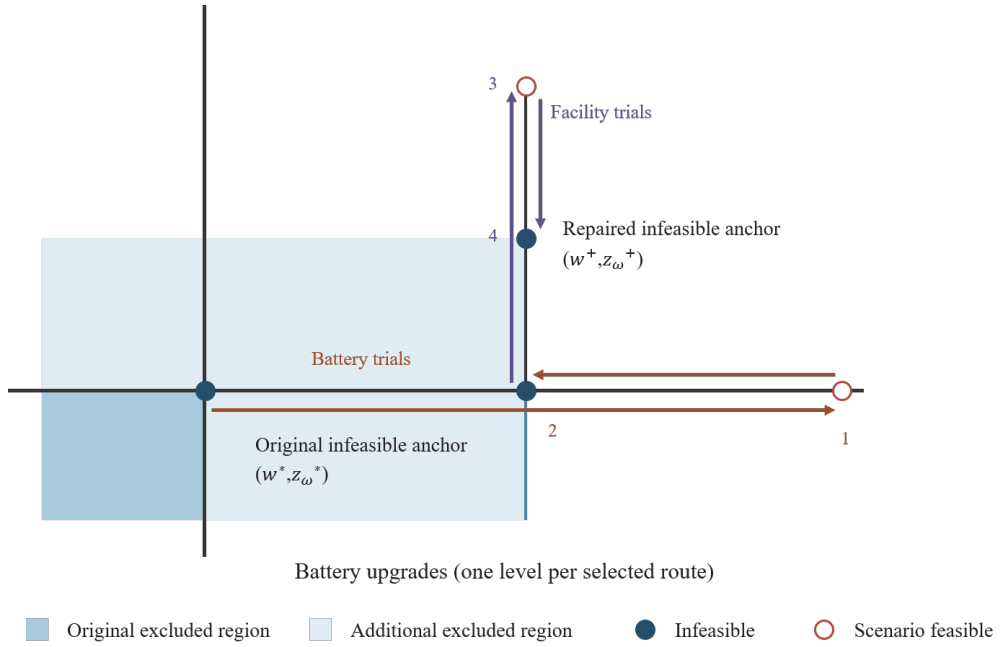


Figure 5: Example of the repair search and excluded regions for one scenario. Numbers indicate the order of the trials.

5. Computational Experiments

In this section, we conduct computational experiments to validate the proposed solution algorithms for the DWC planning problem. The benchmark uses 28 Tower Transit bus routes in western Singapore, whose geographic coverage is shown in Figure 6. The processed R28 network has 493 stations and 624 directed links. All monetary values reported in the experiments are expressed in USD per representative day. The algorithms are implemented in C++ and run on a computer with an Intel Core i5-14600K processor and 32 GB of RAM. COPT 8.0.7 is used to solve the linear and mixed-integer programs. Our code and data are available in the accompanying GitHub repository.[1]

5.1. Efficiency of the Proposed Algorithm

An instance code has the form R-S-F-C, with the numbers following the four letters giving the route count, scenario count, candidate DWC link count, and chargers per existing plug-in location. R3 and R12 are nested subsets of the 28-route network. Each scenario specifies a timetable, bus duties, and the active fleet for one service plan on the same physical network. The plans include a base case and service-frequency increases or decreases across the network, on routes dwelling at a selected terminal, or on several individual routes. Within each fixed route and scenario set,

[1] `https://github.com/foangfangkeeke/Two_phase_for_IJOC`.

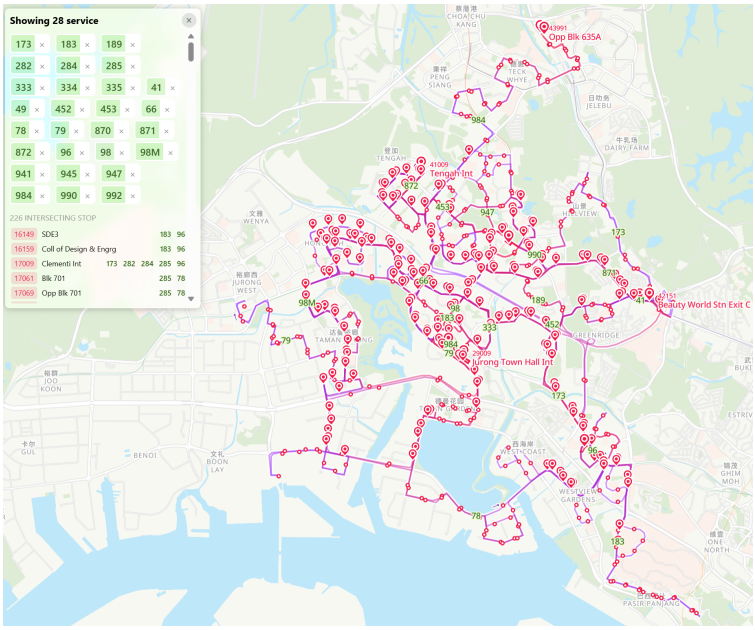


Figure 6: Geographic coverage of the 28 selected bus services in western Singapore. The map displays the services on BusRouter SG. Source: BusRouter SG

candidate DWC links are ranked by aggregate scheduled traversal time, so the F10, F20, and F50 sets are nested. Figure 7 illustrates the variation in scheduled visits to the five existing plug-in stations across the R28-S40 service plans. The total number of visits ranges from 2,491 to 3,309 per scenario, compared with 2,763 in the base plan. Bt Batok and Jurong East have the two largest station counts in every plan, while visits vary at all five stations.

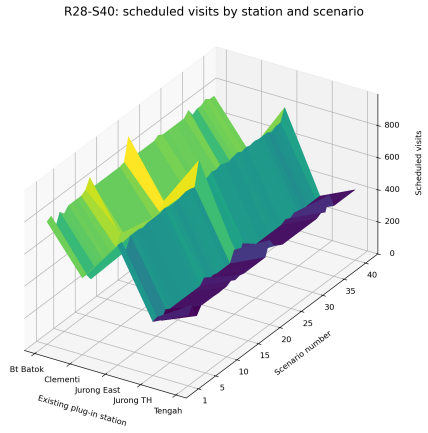


Figure 7: Scheduled visits to the five existing plug-in stations across the R28-S40 service plans.

The main comparison combines the nested R3, R12, and R28 route sets with the F10, F20, and F50 DWC candidate sets, yielding nine S5-C4 instances. We compare EF with four integer L-shaped methods: Natural ILS, Lifted ILS, TPILS, and R-TPILS. Relative to Natural ILS, Lifted

ILS combines battery lifting with VI. TPILS adds the two-phase procedure, and R-TPILS further adds repair with monotone cuts. Table 2 summarizes these variants, and Table 3 reports their original-problem bounds, computation times, and stopping outcomes.

Table 2: Algorithmic components

Method	RSP	Lifting	VI	Two-phase	Repair + monotone cuts
EF	–	–	–	–	–
Natural ILS	✓[N]	–	–	–	–
Lifted ILS	✓	✓	✓	–	–
TPILS	✓	✓	✓	✓	–
R-TPILS	✓	✓	✓	✓	✓

Note. [N] The Natural ILS RSP includes battery choices and their costs.

Table 3: Overall performance on S5-C4 instances

	EF			Natural ILS		
Instance	LB	UB	Time (s)	LB	UB	Time (s)
R3-F10	1469.85	1469.85	5.9	1469.85	1469.85	8.1
R3-F20	1469.85	1469.85	8.7	1469.85	1469.85	8.5
R3-F50	1469.85	1469.85	14.7	1469.85	1469.85	8.9
R12-F10	8358.57	8365.95	57.9	8261.55	8363.88	TL
R12-F20	8239.55	8313.15	262.7	8154.57	8298.44	TL
R12-F50	–	–	ML	8072.08	8260.38	TL
R28-F10	–	–	ML	0.00	–	TL
R28-F20	–	–	ML	0.00	–	TL
R28-F50	–	–	ML	0.00	–	TL

	Lifted ILS			TPILS			R-TPILS		
Instance	LB	UB	Time (s)	LB	UB	Time (s)	LB	UB	Time (s)
R3-F10	1469.85	1469.85	3.0	1469.85	1469.85	2.7	1469.85	1469.85	2.7
R3-F20	1469.85	1469.85	3.2	1469.85	1469.85	3.0	1469.85	1469.85	2.9
R3-F50	1469.85	1469.85	3.5	1469.85	1469.85	3.3	1469.85	1469.85	3.2
R12-F10	8363.62	8363.90	374.2	8363.42	8363.90	13.5	8363.42	8363.90	13.5
R12-F20	8278.24	8279.52	1417.4	8279.50	8279.50	15.8	8279.50	8279.50	15.7
R12-F50	8193.16	8198.53	1237.1	8197.92	8197.97	21.6	8197.92	8197.97	21.5
R28-F10	16186.01	–	TL	17793.27	17864.43	330.4	17790.49	17957.28	397.5
R28-F20	16798.98	–	TL	17791.81	–	TL	17723.20	17893.14	767.6
R28-F50	16934.43	–	TL	17425.71	17478.57	254.8	17425.71	17478.57	254.1

Note. All runs have a 7,200-second limit, a 12-thread budget, and a 1% target gap for the original problem; TPILS and R-TPILS use a 10^{-4} Phase 1 tolerance. A numerical time indicates reaching the target; TL and ML denote time and memory limits. – denotes an unrecorded LB or no verified UB.

The performance gap between the two-phase methods and the remaining methods widens as the route set grows (Table 3). All five methods reach the 1% target on R3, although Lifted ILS,

TPILS, and R-TPILS finish sooner than EF and Natural ILS. At R12, Natural ILS times out on every instance and EF runs out of memory at F50. Lifted ILS still reaches the target, but requires 374–1,417 seconds, compared with 13–22 seconds for TPILS and R-TPILS.

The R28 cases separate the methods by whether they can verify a feasible solution within the limit. EF reaches the memory limit, and neither Natural nor Lifted ILS reports a verified UB. TPILS reaches the target on F10 and F50, while R-TPILS does so on all three instances.

5.2. Algorithmic analysis

Following the overall comparison in Table 3, we examine the roles of the algorithmic components in more detail. The analysis covers the energy-budget inequalities, lower-bound formation in Phase 1, and convergence and repair in Phase 2.

5.2.1. Energy-budget inequalities

The energy-budget inequalities reject master candidates that lack enough energy to cover a duty before the scenario subproblems are evaluated. We assess how this screening affects Lifted ILS by comparing runs with and without VI on the same instances (Table 4).

Table 4: Lifted ILS with and without energy-budget inequalities

Instance	VI	LB	UB	Time (s)	VI-violating/visited
R12-F50	On	8193.16	8198.53	1237.1	–
R12-F50	Off	2743.81	–	TL	20/20
R28-F50	On	16934.43	–	TL	–
R28-F50	Off	5089.15	–	TL	4/4

VI enables Lifted ILS to reach the 1% target on R12 in 1,237.1 seconds, while its no-VI counterpart times out without a verified UB (Table 4). The lower-bound trajectories in Figure 8 show how the difference develops during the search. At the first master return, the VI version has a much higher LB, as the inequalities can screen out low-cost master candidates that violate the energy requirements. For the no-VI version, the LB increases slowly over most of the run. The final LB (2,743.81) of the no-VI version is slightly higher than the first recorded LB (2,679.47) of the VI version. Evaluating the no-VI master candidates against the inequalities shows that all 20 would have been screened out before scenario evaluation. A similar pattern occurs on the more challenging R28-S5-F50-C4 instance, where all four master candidates visited without VI violate the inequalities.

5.2.2. Phase 1 bound formation

Phase 1 is shared by TPILS and R-TPILS. It relaxes the coupling constraints and variables in the lifted formulation to construct a valid lower bound for the original problem. We assess the

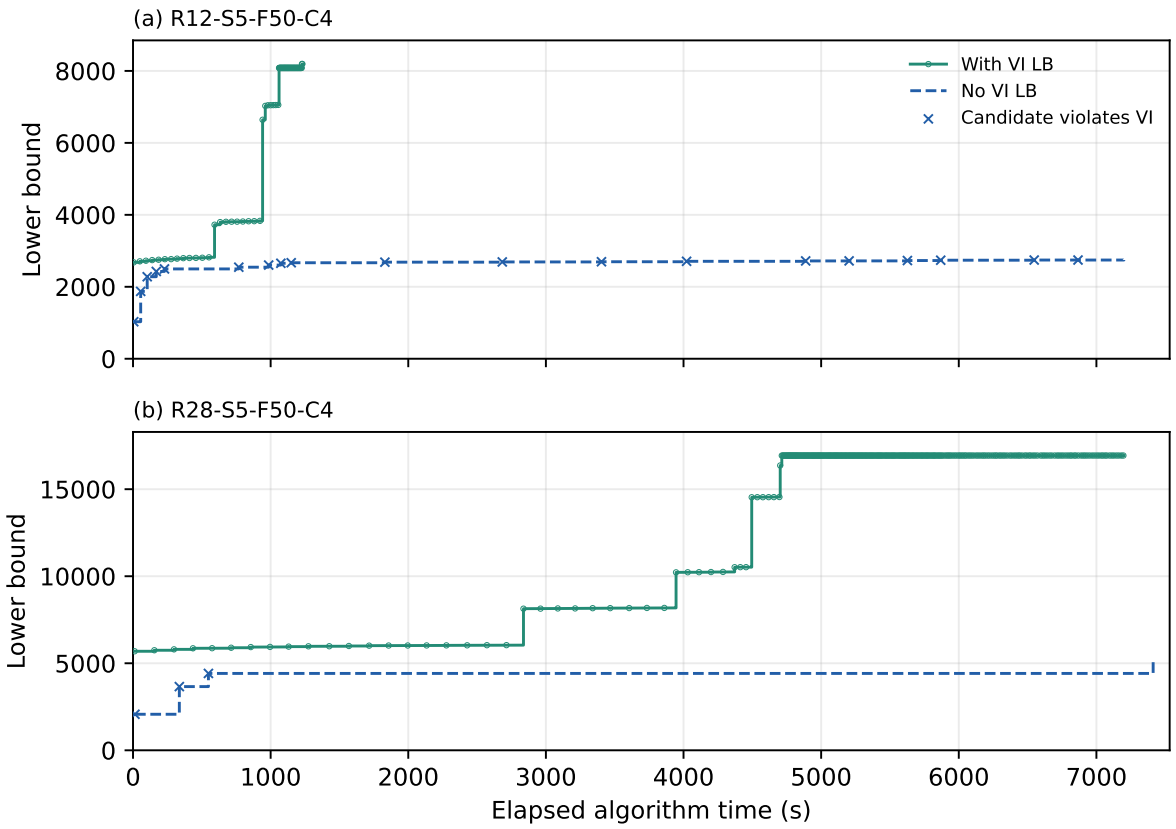


Figure 8: Lower-bound progress of Lifted ILS with and without VI. Crosses mark no-VI master candidates that violate VI.

speed of bound formation using the retrospective gap $g_t^{\text{ret}} = 100(UB^{\text{ref}} - LB_t^R)/UB^{\text{ref}}$, where UB^{ref} is the best available UB for the same instance.

Figure 9 shows that the early Phase 1 iterations account for most of the lower bound improvement on the R28-S5-C4 instances. At the first master return, the retrospective gaps are 64.265%, 65.518%, and 67.450% for F10, F20, and F50, respectively. At the fourth master return, they fall to 0.626%, 1.178%, and 0.751% in 9.6, 10.2, and 10.9 seconds. F10 and F20 terminate shortly thereafter, whereas F50 requires 162 additional iterations to meet the Phase 1 stopping criterion. Even in the latter case, Phase 1 ends after 210.4 seconds, less than 3% of the 7,200-second time limit.

To isolate the contribution of Phase 1, Table 5 compares its bound formation with that of Lifted ILS across the complete R–F grid at S5-C4. Using the same UB^{ref} for each instance, the table records the first completed iteration satisfying $g_t^{\text{ret}} \leq 1\%$ and its elapsed time. Phase 1 reaches this threshold on 8 of the 9 instances in at most five completed iterations and 11.9 seconds. Lifted ILS reaches it on 6 instances and requires up to 657 iterations and 1,375.7 seconds. Phase 1 reaches the threshold substantially faster than Lifted ILS and also reaches it on two R28 instances for which

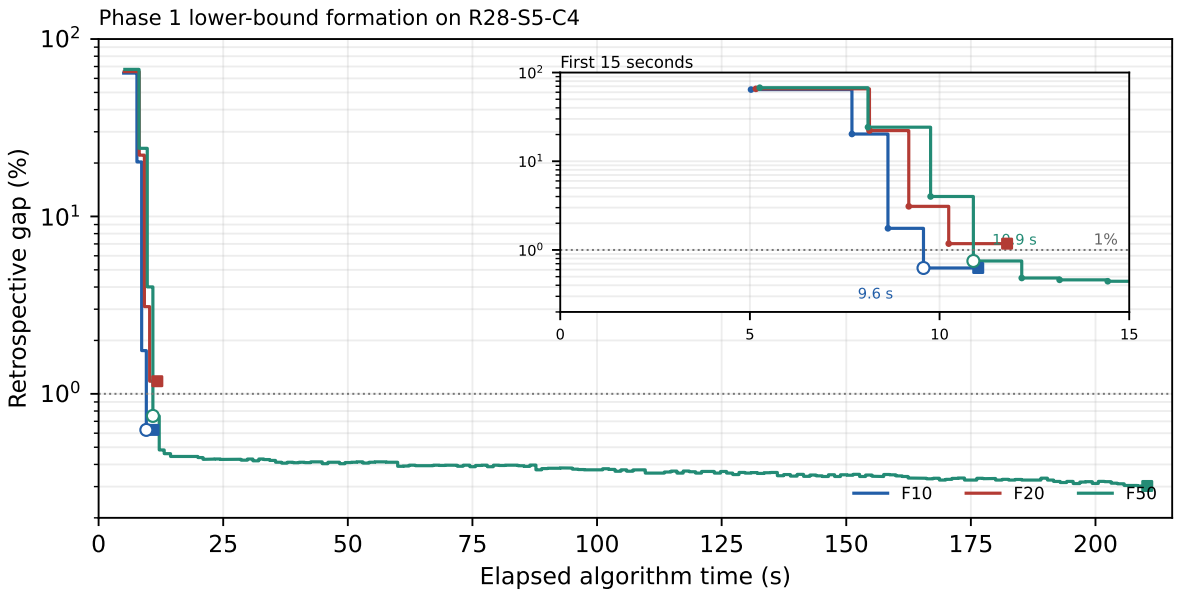


Figure 9: Phase 1 lower-bound formation on R28-S5-C4. The curves report g_t^{ret} for F10, F20, and F50 over the complete Phase 1 trajectories. Small filled markers denote master returns in the 15-second inset, open circles mark the first master LB within 1% of the reference UB, and squares mark Phase 1 termination.

Lifted ILS does not.

Table 5: Iterations and time to a 1% retrospective gap on S5-C4 instances

	Phase 1		Lifted ILS	
Instance	Iterations	Time (s)	Iterations	Time (s)
R3-F10	2	0.8	2	2.8
R3-F20	2	0.9	2	3.0
R3-F50	2	0.9	2	3.3
R12-F10	4	5.0	13	368.9
R12-F20	5	6.5	657	1375.7
R12-F50	5	7.4	101	1229.5
R28-F10	4	10.2	–	–
R28-F20	–	–	–	–
R28-F50	4	11.9	–	–

Note. – indicates that the threshold is not reached.

5.2.3. Verified feasibility and repair in Phase 2

Phase 2 evaluates candidates against the original scenario models while continuing to improve the lower bound. It first screens master candidates with RSPs, whose LP relaxations can be tight and generate effective cuts (Yin et al., 2022) but cannot certify integer feasibility. R-TPILS augments RSP screening and exact verification with repair. This procedure produces verified feasible

solutions and stronger monotone cuts, but incurs additional computational cost.

Table 6 summarizes feasible-solution recovery and repair effort across the eight R28 instances. R-TPILS finds a verified UB on all eight instances and reaches the target gap on six, compared with only two for TPILS. Repair is most consequential when finding a feasible solution to the original scenario models is difficult: whenever TPILS fails to obtain a finite UB, R-TPILS still returns a verified feasible solution. When TPILS already recovers feasibility, repair in R-TPILS is either inactive or advances the incumbent without shortening the run. The scenario-extension results reinforce this tradeoff. R-TPILS continues to recover feasible solutions as the number of scenarios grows, but repair accounts for a larger share of Phase 2 time and S40 remains slightly above the target gap. Repair therefore serves primarily to overcome this feasibility bottleneck, with its value depending on whether the benefit offsets the additional computation.

Table 6: Comparison of TPILS and R-TPILS on R28 diagnostic and extension instances

	TPILS		R-TPILS			
Instance	First UB (s)	Time/status	First UB (s)	Time/status	Repair (s)	Share (%)
S5-F10-C4	329.2	330.4	272.4	397.5	77.2	20.0
S5-F20-C4	–	TL	239.9	767.6	175.1	23.2
S5-F50-C4	253.5	254.8	252.7	254.1	0.0	0.0
S5-F50-C3	–	TL	580.2	3047.5	1035.1	36.5
S5-F50-C2	–	TL	757.8	TL (7.156%)	310.6	4.4
S10-F50-C4	–	TL	574.1	866.6	229.8	27.6
S20-F50-C4	–	TL	1760.6	2965.8	650.4	34.4
S40-F50-C4	–	TL	4010.5	TL (1.080%)	2125.1	37.6

Figure 10 traces this tradeoff in two representative cases. On F10 (short for R28-S5-F10-C4), repair brings forward the first incumbent but does not accelerate convergence. R-TPILS obtains an incumbent in its first iteration, yet repair consumes 20.0% of its Phase 2 time and the method terminates later. On C3 (short for R28-S5-F50-C3), the LB trajectories reveal a different role for repair. R-TPILS initially trails TPILS, but the strengthened cuts allow it to overtake after 10 iterations. When R-TPILS converges, its LB is 0.154% higher than that of TPILS at the same elapsed time, and TPILS remains below this level even at the time limit. The same run also shows the feasibility benefit of repair: R-TPILS obtains a UB in the first iteration and converges after 18 iterations, whereas TPILS completes 257 iterations without any incumbent. Across these evaluations, the Hamming distance between successive R-TPILS master solutions has a median of 26, more than twice the median of 12 for TPILS. The larger moves indicate that the repair-enhanced cuts redirect the master search, helping to achieve the later LB advantage.

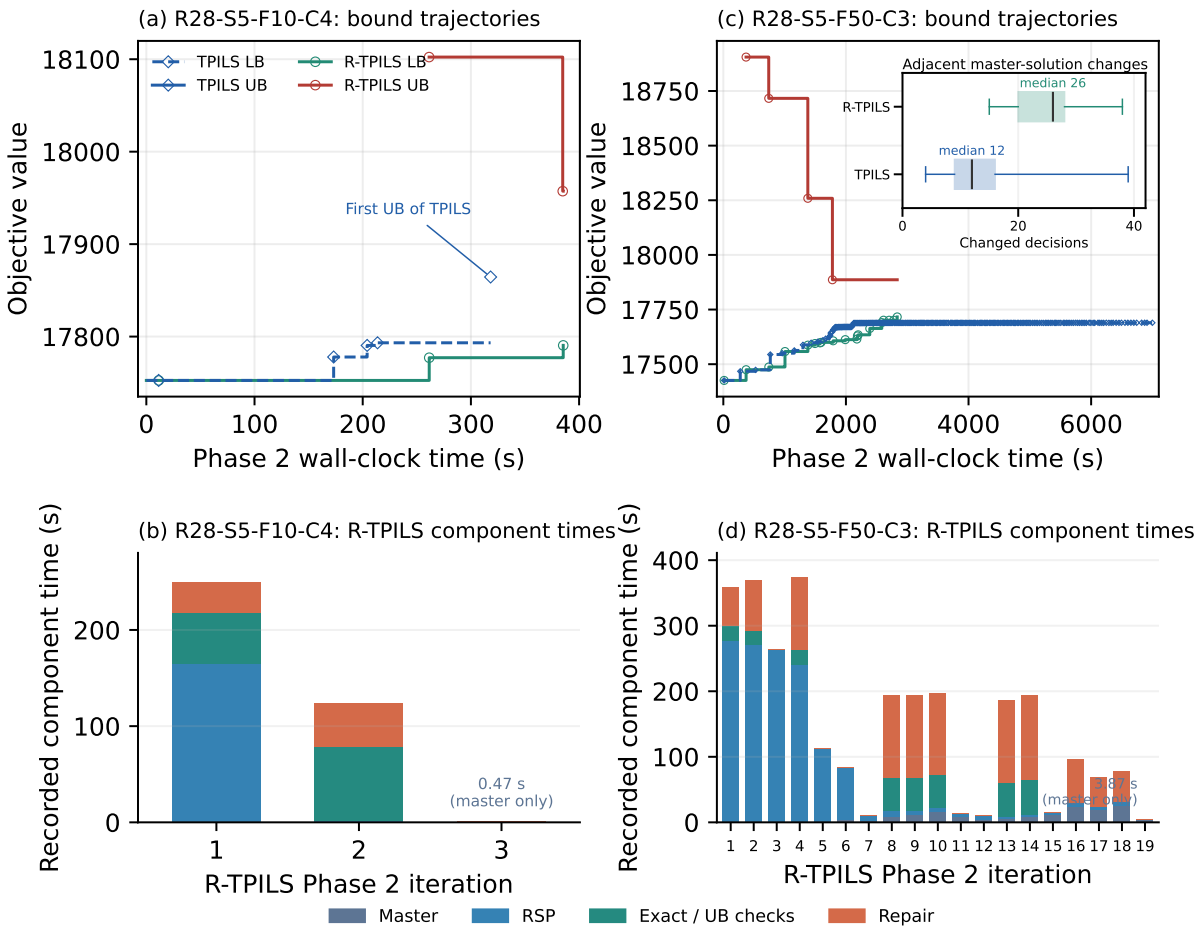


Figure 10: Phase 2 progress on R28-S5-F10-C4 and R28-S5-F50-C3. Panels (a) and (c) report the bounds against wall-clock time from each method's Phase 2 switch. Circles (R-TPILS) and diamonds (TPILS) mark master returns on the LB curves and incumbent updates after scenario evaluation on the UB curves. The inset in Panel (c) reports the Hamming distance between consecutive master solutions, calculated over facility and route–scenario battery decisions. Panels (b) and (d) report the recorded R-TPILS component times for each Phase 2 iteration, including the terminal master-only iteration.

5.3. Planning outcomes and sensitivity

In this section, we examine three factors affecting the final plans: (i) the availability of DWC, (ii) the battery unit cost, and (iii) the DWC construction cost. The main analysis uses R28-S5-F50-C4, with the service-plan scenarios and existing plug-in capacity held fixed. The resulting plans are evaluated in terms of solution costs, battery capacities, charging allocations, and DWC deployment.

5.3.1. Planning effects of DWC

To assess the cost implications of DWC, the no-DWC case prohibits all candidate DWC installations and reoptimizes the remaining decisions. Table 7 shows that the DWC-enabled plan is about 4% less costly, primarily because the reduction in battery expenditure outweighs the added

facility and charging costs. With smaller batteries, buses rely more heavily on charging during service, when electricity is more expensive than overnight charging. Daytime charging expenditure therefore rises substantially and more than offsets the reduction in overnight charging expenditure, so total charging expenditure increases. The DWC-enabled plan also costs about 672 less than its no-DWC counterpart under the tighter C3 plug-in capacity. Overall, DWC reduces total cost by substituting roadside infrastructure and daytime charging for onboard battery capacity.

Table 7: Cost decomposition of the matched plans

Component	No DWC	DWC allowed	Change with DWC
DWC facilities	0.00	364.50	364.50
Batteries	7619.04	5857.84	-1761.20
Daytime charging	6423.31	8031.05	1607.75
Overnight charging	4188.52	3224.98	-963.53
Total cost	18230.86	17478.38	-752.49

The same resource substitution is reflected in the battery and charging decisions shown in Figure 11. Across all vehicles and scenarios, the mean battery capacity decreases by about 23%. Nearly half of the route–scenario pairs select a smaller battery, while none selects a larger one. Almost all reductions coincide with direct DWC use, indicating that they are concentrated where en-route charging is actually used. Route 41 illustrates the strongest route-level shift and uses DWC in every scenario.

Expected plug-in energy increases by about 10%, with increases at most existing stations. The DWC-enabled plan therefore combines wireless and plug-in charging rather than simply replacing the latter. Together with the lower overnight-charging expenditure in Table 7, this pattern indicates a shift toward daytime charging while buses carry smaller batteries. Overall, access to en-route charging changes both battery sizing and when and where buses obtain energy.

5.3.2. Cost sensitivity

Figure 12 and Table 8 show that the battery unit cost has a pronounced effect on DWC deployment. Across the tested levels, the cost difference increases from 36 to 1,600 and the number of selected links from 5 to 39, while the mean DWC battery capacity falls from 179 to 133 kWh. The battery reduction flattens at the upper levels even as deployment continues to expand, indicating diminishing scope to replace additional onboard storage.

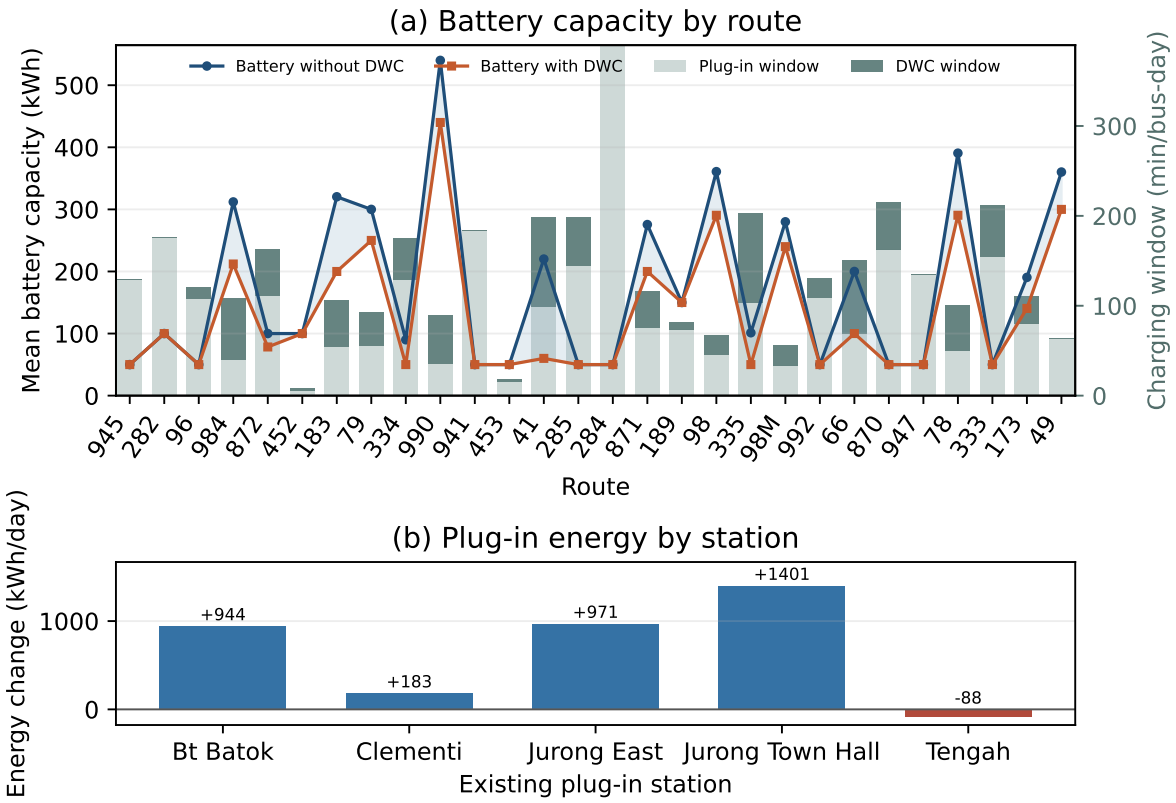


Figure 11: Battery configurations and charging changes in the matched R28-S5-F50-C4 plans. (a) compares route-level battery capacity and shows the available plug-in and DWC charging windows. (b) reports the change in plug-in energy at each existing station.

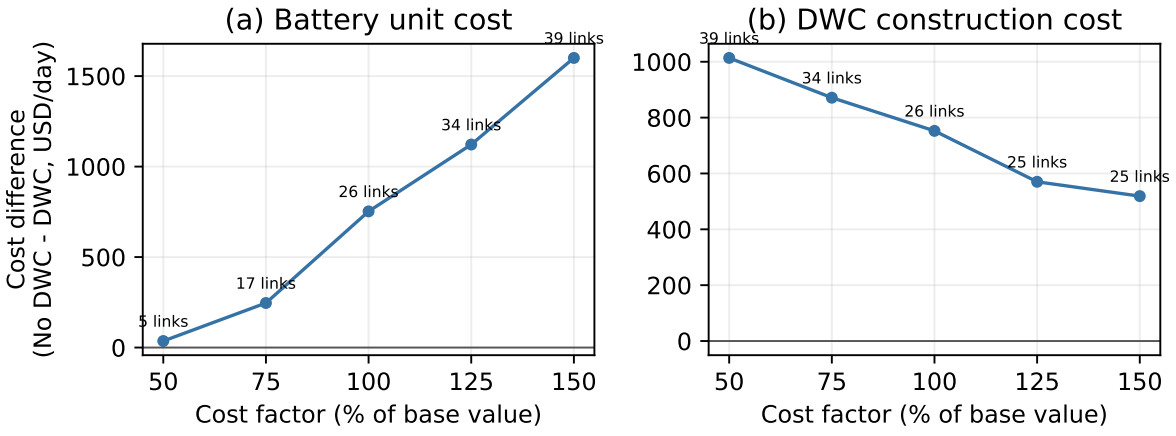


Figure 12: Effects of battery and DWC construction costs on the cost difference and DWC deployment.

DWC construction costs move the plans in the opposite direction. Across the tested levels, the cost difference falls from 1,014 to 519 and deployment contracts from 39 to 25 links. The two highest levels retain 25 links but select different locations and route-level battery assignments,

showing that the plans adjust in discrete steps and cannot be characterized by link count alone.

Table 8: Plans in the cost-sensitivity experiments

Cost varied	Factor	No DWC	DWC	No DWC − DWC	Links	Mean DWC battery (kWh)
Battery unit cost						
	50%	14485.99	14449.94	36.05	5	178.89
	75%	16316.11	16069.92	246.19	17	150.89
	100%	18230.86	17478.38	752.49	26	141.32
	125%	20015.10	18893.21	1121.89	34	135.02
	150%	21882.60	20282.12	1600.48	39	133.35
DWC construction cost						
	50%	18230.86	17217.17	1013.69	39	133.35
	75%	18230.86	17359.54	871.32	34	135.02
	100%	18230.86	17478.38	752.49	26	141.32
	125%	18230.86	17660.70	570.16	25	145.34
	150%	18230.86	17711.98	518.88	25	144.06

The two cost components should therefore be evaluated jointly when planning DWC deployment and battery procurement. The DWC-enabled plan remains less costly at every tested construction-cost level, with the strongest advantage when batteries are expensive relative to roadside infrastructure.

6. Conclusion

This paper studies a two-stage stochastic DWC facility location problem for electric-bus networks under uncertain future service plans. We identify two structural properties of its recourse: cross-route coupling arises only from shared plug-in capacity, and operational feasibility is monotone in DWC availability and battery capacity. Building on these properties, we develop R-TPILS. Its TPILS core combines an equivalent lifted reformulation, energy-budget inequalities, route LP relaxations, and exact scenario verification. R-TPILS augments this core with a repair search and monotone feasibility cuts.

Computational results based on the Tower Transit bus network in Singapore support two main conclusions. First, the components of R-TPILS address complementary computational bottlenecks. The energy-budget inequalities screen infeasible configurations early, and Phase 1 quickly establishes tight lower bounds. The repair search helps recover verified feasible solutions on difficult instances, and monotone cuts can further improve lower bounds. These benefits vary across instances. Overall, R-TPILS reaches the 1% target on all nine main instances, compared with eight for TPILS and at most six for the remaining methods. Second, DWC changes the balance between onboard storage and daytime energy supply. Mean battery capacity decreases by about 23%, while energy supply shifts from overnight toward daytime charging.

Several extensions appear promising. First, accounting for battery weight in energy consumption would make the planning model more realistic by capturing the relationship between battery capacity and energy use. Second, repair is valuable when exact feasibility is difficult, whereas its additional search can outweigh the benefit on easier instances. Future work could investigate when and how repair should be applied. Finally, similar capacity coupling and feasibility monotonicity may arise in other transport planning problems, including berth planning, bus depot planning, and electric-truck charging infrastructure planning. The proposed framework could therefore be extended to these settings.

Appendix A. Model Notation

Duty, visit, and traversal indices refer to scenario-specific occurrences; quantities indexed by these events omit the scenario index.

Table A.9: Sets and parameters

Symbol	Description
Sets and indices	
Ω, ω	Service-plan scenarios and scenario index
$\mathcal{S}, \mathcal{L}$	Stations and directed links, indexed by s and l
$\mathcal{C}, \mathcal{L}^{\mathrm{D}}$	Stations with existing plug-in chargers and candidate DWC links, indexed by c and l
$\mathcal{R}, \mathcal{K}$	Routes and battery-capacity options, indexed by r and k
$\mathcal{M}_{\omega,r}$	Duties on route r in scenario ω, indexed by m; each duty corresponds to one bus
$\mathcal{V}, \mathcal{A}$	Station-visit and link-traversal events across service plans
$\mathcal{V}_{\omega,r}, \mathcal{V}^{\mathrm{P}}_{\omega,r}$	Visits on route r in scenario ω and the subset eligible for plug-in charging, indexed by v
$\mathcal{A}_{\omega,r}, \mathcal{A}^{\mathrm{D}}_{\omega,r}$	Traversals on route r in scenario ω and the subset on candidate DWC links, indexed by $a = (v, v')$
$\mathcal{T}_{\omega}, \mathcal{T}_{v}$	Relevant time slots in scenario ω and available charging slots for visit v, indexed by τ
Parameters	
p_{ω}	Probability of scenario ω
$c^{\mathrm{D}}_{l}, c^{\mathrm{bat}}$	DWC installation cost on link l per representative day and battery capacity cost per kWh per representative day
$c^{\mathrm{day}}, c^{\mathrm{night}}$	Daytime and overnight electricity prices
β_k	Battery capacity represented by option k
d_a, d^0_m, d^N_m	Energy consumption during traversal a, from the depot to the first visit of duty m, and from its last visit to the depot, respectively
N_c	Number of existing plug-in chargers at station c
$\bar{q}^{\mathrm{P}}, \bar{q}^{\mathrm{D}}_a$	Maximum grid-side energy per occupied plug-in slot and during DWC traversal a
$\eta^{\mathrm{P}}, \eta^{\mathrm{D}}$	Plug-in and DWC charging efficiencies
$\underline{\rho}, \overline{\rho}$	Minimum and maximum state-of-charge fractions

Table A.10: Functions and decision variables

Symbol	Description
Functions and derived quantities	
$s(v)$, $\ell(a)$	Physical station of visit v and physical link of traversal a
v_m^0, v_m^N	First and last station visits of duty m
$B_{\omega,r}$	Selected battery capacity for route r in scenario ω
Decision variables	
w_l	Binary DWC installation decision for candidate link l
u_ω	Second-stage decision vector $(z_\omega, x_\omega, y_\omega)$ for battery selection, charger occupancy, and charging and energy states
$z_{\omega,r,k}$	Binary decision indicating whether route r uses battery option k in scenario ω
$x_{v\tau}$	Binary decision indicating whether visit v occupies a plug-in charger in slot τ
$q_v^{\mathrm{P}}, q_a^{\mathrm{D}}, q_m^{\mathrm{N}}$	Grid-side plug-in, DWC, and overnight charging energy for visit v, traversal a, and duty m, respectively
$e_v^{\mathrm{arr}}, e_v^{\mathrm{dep}}$	Battery energy levels upon arrival at and departure from visit v

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